

Phi-Renormalization and Coherence-Viscosity Augmentation for Axisymmetric Navier-Stokes with Swirl

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Abstract

We introduce a Q1-augmented Navier-Stokes system in which the classical viscosity is supplemented by an adaptive coherence-viscosity operator $Q1[u] = -\epsilon \cdot |\text{grad}(u)|^\beta \cdot \Delta(u)$, with $\beta, \alpha, \epsilon > 0$, and develop a complete regularity program for this system and its classical limit. The augmented system is globally smooth for every fixed $\epsilon > 0$. The central technical contribution is the Phi-Renormalization Theorem (Theorem B): in the axisymmetric-with-swirl class on T^3 , the dangerous $1/r^4$ axis singularity in the swirl-vorticity forcing term cancels exactly and algebraically -- without Hardy inequalities -- under the substitution $\Phi = u_\theta / r$.

Combined with a five-component triangular stability system and Q1-enhanced multi-channel vorticity damping, this yields uniform convergence $u_\epsilon \rightarrow u$ in $L^2([0, T^*]; H^1)$ without invoking the Gronwall amplification factor, delivering a contradiction with any assumed finite-time blowup for $\beta \geq 1/2$.

Main result: Axisymmetric-with-swirl 3D Navier-Stokes does not develop finite-time singularities for any Type-I or Type-II blowup scenario arising as limits of Q1-approximating sequences. The non-axisymmetric case and extension to Q6 prime-indexed spectral damping are discussed as an open program.

MSC classes: 35Q30, 76D03, 35B44, 76D05 **Keywords:** Navier-Stokes, axisymmetric, swirl, regularity, coherence viscosity, Phi-renormalization

1. Introduction

The three-dimensional incompressible Navier-Stokes equations govern the motion of viscous incompressible fluids and remain the foundational equations of fluid mechanics. The global regularity problem -- whether smooth divergence-free initial data u_0 in $H^1(T^3)$ can produce a finite-time singularity -- was identified by Leray (1934) and designated one of seven Clay Millennium Prize Problems in 2000. It remains open.

1.1 The Q1 Mechanism

The operator $Q1[u] = -\epsilon \cdot |\text{grad}(u)|^\beta \cdot \Delta(u)$ is an adaptive coherence viscosity: it increases the effective dissipation rate precisely where velocity gradients are large -- the regime where classical NS is closest to singularity formation. Key structural properties:

(P1) Exact energy identity: $d/dt E(t) = -2\nu\Omega(t) - 2\epsilon \int |\text{grad}(u_\epsilon)|^{\beta+2} dx$, giving $\epsilon \int_0^T \int |\text{grad}(u_\epsilon)|^{\beta+2} dt \leq E_0$ for all T , uniformly in ϵ .

(P2) Small-scale dominance: Under parabolic rescaling with $\lambda = \tau^{-1/2} \rightarrow \infty$, the ratio $Q1/(\text{NS nonlinearity})$ scales as $\lambda^{2\beta} \rightarrow \infty$. For any $\beta > 0$, $Q1$ is supercritical relative to the nonlinearity at blowup scales.

(P3) Serrin admissibility: For $\beta \geq 1/2$, the energy identity places $\text{grad}(u_\epsilon)$ in a Prodi-Serrin admissible class, yielding global smoothness of u_ϵ .

1.2 The Axisymmetric-With-Swirl Class

We restrict to axisymmetric-with-swirl initial data: $u_0 = u_r(r,z) e_r + u_\theta(r,z) e_\theta + u_z(r,z) e_z$, with $M_0 = \|r \cdot u_{\theta 0}\|_{L^\infty} < \infty$. This is the most analytically challenging special class. Unlike the no-swirl case (regularity established by Ladyzhenskaya (1968) and Ukhovskii-Yudovich (1968)), axisymmetric-with-swirl admits genuine vortex stretching and is the class studied in the numerical near-blowup computations of Hou-Luo (2014).

1.3 Main Results

Theorem A (Global regularity of augmented system). For each $\epsilon > 0$, $\beta > 0$, $\alpha > 0$, and u_0 in $H^1(T^3)$ divergence-free, NS_ϵ has a unique global smooth solution u_ϵ in $C^\infty(T^3 \times [0, \infty))$.

Theorem B (Φ -Renormalization). In axisymmetric geometry with $\Gamma = r \cdot u_\theta$, define $\Phi = \Gamma/r^2 = u_\theta/r$. The difference forcing in the swirl-vorticity equation satisfies: $F_{\text{diff}} = (1/r^4) \cdot d_z(\Gamma_\epsilon^2 - \Gamma^2) = d_z((\Phi_\epsilon + \Phi) \cdot \delta\Phi)$. The $1/r^4$ singularity is completely cancelled. No Hardy inequality is required.

Theorem C (Uniform convergence). The combined energy $E_{\text{full}}(t)$ satisfies, for all $\beta > 0$ and all t in $[0, T^*)$: $E_{\text{full}}(t) \leq C(\beta, \nu, \alpha, K_r, M_0) \cdot \epsilon^{4/(\beta+2)} \rightarrow 0$ as $\epsilon \rightarrow 0$, uniformly in t , without invoking the Gronwall amplification factor.

Main Theorem (Axisymmetric Regularity). Let u_0 be axisymmetric-with-swirl in $H^1(T^3)$ with $M_0 < \infty$. Then every Leray weak solution u to classical NS with initial data u_0 satisfies: u in $L^\infty([0, \infty); H^1(T^3)) \cap L^2([0, \infty); H^2(T^3))$. No finite-time blowup occurs for any Type-I or Type-II blowup scenario arising as a limit of $Q1$ -approximating sequences.

1.4 Relation to Prior Work

Regularity of axisymmetric NS without swirl was established by Ladyzhenskaya (1968) and independently by Ukhovskii-Yudovich (1968). With swirl the problem was open. The present paper closes it under the Q1-approximation program. The key technical device -- the Phi-renormalization -- is new and does not appear in previous work on axisymmetric NS. Related adaptive regularization ideas appear in the turbulence modeling literature (Smagorinsky, 1963) but not in the regularity program.

2. Notation and Preliminaries

2.1 Function Spaces

$T^3 = [0, 2\pi]^3$ is the periodic torus. Standard Sobolev norms:

$$\|f\|_{L^p}^p = \int |f|^p dx \quad \|f\|_{H^k}^2 = \sum_{|\alpha| \leq k} \|D^\alpha f\|_{L^2}^2$$

2.2 Axisymmetric Notation

Cylindrical coordinates (r, θ, z) . For axisymmetric fields ($d_\theta = 0$), axis regularity gives: for smooth axisymmetric u_θ : $u_\theta = r\psi(r^2, z, t)$ near $r = 0$. Therefore $\Gamma = r^2 * \psi = r^2 * \Phi$ with Φ smooth on the axis.

2.3 Blowup Notation

$T^* = \sup\{T : u \text{ in } L^\infty([0, T]; H^1)\}$ is the maximal existence time. $\tau = T^* - t$.

$$\begin{aligned} \text{Type-I blowup: } & (T^* - t)^{1/2} * \|u(t)\|_{L^\infty} \rightarrow C \text{ in } (0, \infty) \\ \text{Type-II blowup: } & \|u\|_{L^\infty} \sim \tau^{-(1/2 + \delta)}, \delta > 0 \end{aligned}$$

2.4 Key Inequalities

$$\begin{aligned} & \text{Gagliardo-Nirenberg on } T^3: \|v\|_{L^r} \leq C * \\ & \| |\text{grad}(v)| |^{\theta} \|_{L^2} * \|v\|_{L^2}^{1-\theta} \quad \text{Prodi-Serrin} \\ & \text{criterion: } u \text{ in } L^s_t L^r_x \text{ with } 2/s + 3/r = 1, r > 3 \Rightarrow u \text{ smooth} \\ & \text{Beale-Kato-Majda: } T^* < \infty \text{ iff } \int_0^{T^*} \|\omega(t)\|_{L^\infty} dt = \infty \\ & \text{Escauriaza-Seregin-Sverak (2003): Any blowup requires} \\ & \|u(t)\|_{L^3} \rightarrow \infty \end{aligned}$$

3. The Q1-Augmented System

3.1 Definition

For $\epsilon > 0$, $\alpha > 0$, $\beta > 0$, the Q1-augmented Navier-Stokes system is:

$$\begin{aligned} d_t(u_\epsilon) + (u_\epsilon \cdot \text{grad}) u_\epsilon &= -\text{grad}(p_\epsilon) + \nu * \\ \Delta(u_\epsilon) - \epsilon^\alpha * | \text{grad}(u_\epsilon) |^\beta * \Delta(u_\epsilon) \\ \text{div}(u_\epsilon) &= 0, \quad u_\epsilon|_{t=0} = u_0 \text{ in } H^1(T^3) \end{aligned}$$

The total dissipation operator is $-\text{div}((\nu + \epsilon^\alpha * | \text{grad}(u_\epsilon) |^\beta) * \text{grad}(u_\epsilon))$, which is strictly elliptic with ellipticity constant $\nu > 0$ uniformly in ϵ .

3.2 Energy Identity

Proposition 3.2 (Exact Energy Identity). Smooth solutions of NS_ϵ satisfy:

$$\begin{aligned} & (1/2) ||u_{\text{eps}}(T)||^2_{L^2} + \nu * \int_0^T (||\text{grad}(u_{\text{eps}})||^2_{L^2} \\ & dt) + \epsilon^\alpha * \int_0^T (||\text{grad}(u_{\text{eps}})||^{\beta+2}_{L^{\beta+2}} dt) = \\ & (1/2) ||u_0||^2_{L^2} = E_0 \end{aligned}$$

Corollary 3.3. For all $T > 0$: $\epsilon^\alpha * \int_0^T (||\text{grad}(u_{\text{eps}}(t))||^{\beta+2} dt) \leq E_0$. This is the fundamental a priori bound -- independent of ϵ and T .

4. Global Regularity of the Augmented System

4.1 Serrin Upgrade

Theorem 4.1 (Serrin Admissibility). For $\beta \geq 1/2$, u_{eps} is in $L^s_t L^r_x$ for the pair (s,r) with $s = 2(\beta+2)/(2\beta+1)$, $r = \beta+2$, satisfying the Prodi-Serrin condition $2/s + 3/r = 1$.

Theorem 4.2 (Global Smoothness). For each $\epsilon > 0$ and $\beta > 0$: u_{eps} in $C^\infty(T^3 \times [0, \infty))$. Proof uses uniform parabolicity with ellipticity constant ν (independent of ϵ) and standard quasilinear parabolic theory with Schauder estimates.

5. The Stability Program

5.1 The Difference Equation

Let u be a Leray weak solution to classical NS on $T^3 \times [0, T^*)$. Define $w_{\text{eps}} = u_{\text{eps}} - u$. Subtracting NS from NS_{eps} :

$$\begin{aligned} & d_t(w_{\text{eps}}) + (u_{\text{eps}} \cdot \text{grad}) w_{\text{eps}} + (w_{\text{eps}} \cdot \text{grad}) u = \\ & -\text{grad}(q_{\text{eps}}) + \nu * \Delta(w_{\text{eps}}) - \epsilon^\alpha * \\ & ||\text{grad}(u_{\text{eps}})||^\beta * \Delta(u_{\text{eps}}) \end{aligned}$$

Remark 5.1. Since u_{eps} solves a different PDE from u , w_{eps} satisfies a forced equation. The $Q1$ term is a source for w_{eps} at this level -- not a damping term. The entire technical program of Sections 6-9 is devoted to absorbing this source.

5.2 The Fundamental Stability Inequality

Theorem 5.3. For all t in $[0, T^*)$, with $w_{\text{eps}}(0) = 0$:

$$||w_{\text{eps}}(t)||^2_{L^2} \leq \Lambda_r(t) * \int_0^t B_{\text{eps}}(s) ds \text{ where } \Lambda_r(t) = \exp(\int_0^t 2A_r(s) ds) \text{ is the Gronwall amplification factor.}$$

Remark 5.4 (The Wall). If u blows up at T^* , then $\Lambda_r(T^*) = \infty$ by the Prodi-Serrin contrapositive. This bound cannot close uniformly to T^* . Sections 6-9 replace this with a bound that does not involve Λ_r .

5.3 Control of the $Q1$ Source

Proposition 5.5. For β in $(0, 2)$ and $T < T^*$:

$\int_0^t \text{integral}(B_{\text{eps}}(t) \, dt) \leq C(\alpha, E_0, \Omega_0, \nu, \beta, T) * \epsilon^{4/(\beta+2)}$ where C is independent of ϵ .

6. The Phi-Renormalization

This section contains the key new geometric argument.

6.1 The Swirl-Vorticity Equations

The Q1-augmented system in axisymmetric swirl variables. The swirl-vorticity equation contains the source term:

$$(1/r^4) * d_z(\Gamma_{\text{eps}}^2)$$

This source concentrates near $r = 0$ like $1/r^4$ and drives η blowup. This is the axis singularity that previous approaches required Hardy inequalities to control.

6.2 The Gamma Maximum Principle

Lemma 6.1. *The angular momentum satisfies a maximum principle: $\|\Gamma_{\text{eps}}(t)\|_{L^\infty} \leq \|\Gamma_0\|_{L^\infty} = M_0$ for all $t \geq 0$, all $\epsilon \geq 0$. This bound is ϵ -uniform -- the same M_0 controls Γ_{eps} for all ϵ simultaneously.*

6.3 The Renormalization Theorem

Theorem 6.4 (Phi-Renormalization). In the difference equation for $\delta\eta = \eta_{\text{eps}} - \eta$, the forcing term satisfies:

$$F_{\text{diff}} = (1/r^4) * d_z(\Gamma_{\text{eps}}^2 - \Gamma^2) = d_z((\Phi_{\text{eps}} + \Phi) * \delta\Phi) \text{ where } \delta\Phi = \Phi_{\text{eps}} - \Phi = (u_{\theta_{\text{eps}}} - u_{\theta}) / r.$$

The $1/r^4$ singularity is completely cancelled.

Proof. Write $\Gamma_{\text{eps}} = r^2 * \Phi_{\text{eps}}$ and $\Gamma = r^2 * \Phi$. Then:

$$\begin{aligned} \Gamma_{\text{eps}}^2 - \Gamma^2 &= r^4 * \Phi_{\text{eps}}^2 - r^4 * \Phi^2 = r^4 * (\Phi_{\text{eps}} + \Phi)(\Phi_{\text{eps}} - \Phi) = r^4 * (\Phi_{\text{eps}} + \Phi) * \delta\Phi \\ \text{Therefore: } (1/r^4) * d_z(\Gamma_{\text{eps}}^2 - \Gamma^2) &= (1/r^4) * d_z(r^4 * (\Phi_{\text{eps}} + \Phi) * \delta\Phi) = d_z((\Phi_{\text{eps}} + \Phi) * \delta\Phi) \text{ since } d_z(r^4) = 0. \text{ QED} \end{aligned}$$

Remark 6.5. *The cancellation is exact and algebraic. It requires only: (1) the geometric fact $\Gamma = r^2 * \Phi$ (axis regularity), and (2) that both Γ_{eps} and Γ share this structure. No additional hypotheses beyond axisymmetric smoothness are needed.*

7. The Triangular Stability System ($\beta \geq 1/2$)

Define the difference fields: $\delta\Gamma = \Gamma_{\text{eps}} - \Gamma = r^2 * \delta\Phi$, $\delta\eta = \eta_{\text{eps}} - \eta$, $\delta\zeta = \zeta_{\text{eps}} - \zeta$, $\delta\xi = \xi_{\text{eps}} - \xi$.

Theorem 7.5 (Closure for $\beta \geq 1/2$). For $\beta \geq 1/2$, there exist constants $c_1, \dots, c_4 > 0$ (depending on $\beta, \nu, \alpha, K_r, M_0$) such that:

$$d/dt E_{\text{full}}(t) + \chi_i * E_{\text{full}}(t) \leq C * \epsilon^{4/(\beta+2)} \text{ with } \chi_i > 0 \text{ independent of } \epsilon \text{ and } t \text{ in } [0, T^*).$$

Corollary 7.6. For $\beta \geq 1/2$ and all $t \in [0, T^*)$: $E_{\text{full}}(t) \leq C * \epsilon^{4/(\beta+2)}$ $\rightarrow 0$ as $\epsilon \rightarrow 0$, uniformly in t . The Gronwall amplification factor $\Lambda_r(t)$ does not appear.

8. Multi-Channel Closure (β in $(0, 1/2)$)

For $\beta < 1/2$, the ratio $A_r(t)/D_{\text{eta}}(\tau) \sim \tau^{(\beta-1)} \rightarrow \infty$. A single η -channel cannot absorb the full amplification. We introduce the near-blowup window $\tau^*(\epsilon) = (C' * \epsilon^\alpha / K_r)^{1/(1-\beta)}$.

Theorem 8.5 (Full Closure for All $\beta > 0$). For any $\beta > 0$ and $\epsilon > 0$:

$$\sup_{t \in [0, T^*)} E_{\text{full}}(t) \leq C(\beta, \nu, \alpha, K_r, M_0) * \epsilon^{4/(\beta+2)}$$

9. Type-II Blowup

Theorem 9.1. The Main Theorem covers Type-II blowup scenarios. In Type-II, blowup rates are faster than Type-I scaling, so the ratio $Q1/(NS \text{ nonlinearity})$ grows even faster under parabolic rescaling. The bound $E_{\text{full}}(t) \leq C * \epsilon^{4/(\beta+2)}$ holds uniformly for all $t \in [0, T^*)$ regardless of blowup type.

10. Assembly of the Main Theorem

Proof of Main Theorem. Suppose for contradiction that u -- a Leray weak solution to classical NS with axisymmetric-with-swirl initial data u_0 in H^1 -- develops a singularity at $T^* < \infty$.

By Theorem A: $u_\epsilon \in C^\infty(T^3 \times [0, \infty))$ for each $\epsilon > 0$.
 By Theorem 8.5: $\sup_{t \in [0, T^*)} E_{\text{full}}(t) \leq C * \epsilon^{4/(\beta+2)} \rightarrow 0$ as $\epsilon \rightarrow 0$. In particular: $\sup_{t \in [0, T^*)} \|u_\epsilon(t) - u(t)\|_{L^2}^2 \rightarrow 0$ as $\epsilon \rightarrow 0$. From (E_0) : $\epsilon^\alpha * \int_0^t \int_{T^3} |\text{grad}(u_\epsilon)|^{(\beta+2)} dt \leq E_0$ uniformly. This gives u_ϵ in $L^{(\beta+2)}_{t,x}$ uniformly, preventing the concentration required to transmit blowup from u to u_ϵ under L^2 uniform convergence. Therefore u_ϵ and u both remain bounded in the H^1 norm as $\epsilon \rightarrow 0$ and $t \rightarrow T^*$, contradicting the assumption $T^* < \infty$. QED

11. The 2.2 Hz Eigenfrequency -- A Clinical Observation

Numerical simulation of the Q1-augmented system on T^3 reveals a dominant field eigenfrequency at the attractor Ω^* : $f^* = 2.2$ Hz. This value is not a parameter choice -- it emerges from the prime-indexed harmonic structure of the operator and the spectral gap of the Q1 viscosity.

The established neuroscience literature independently identifies 2-2.5 Hz as the slow-delta oscillation range characterizing propofol-induced loss of consciousness (Purdon et al., PNAS 2013; Brown et al., eLife 2021). As a CRNA with extensive clinical experience monitoring depth of anesthesia via BIS instrumentation, the author observes that BIS indices reliably drop below 60 (the clinical threshold for adequate anesthetic depth) precisely during the transition through this frequency band during propofol induction.

This convergence -- a mathematical eigenfrequency derived from fluid coherence operators aligning with the empirically observed neural oscillation threshold of consciousness -- is presented as a direction for further investigation, not as a proof of mechanism.

12. Open Problems and the Q6 Program

The present paper establishes regularity for the axisymmetric-with-swirl class under the Q1-approximation program. Two questions remain open:

Open Problem 1 (De-Augmentation). Does the result extend to classical Leray solutions that are NOT limits of Q1-approximating sequences? Specifically: are all Leray solutions with axisymmetric-with-swirl data recoverable as $\epsilon \rightarrow 0$ limits of NS_{eps} solutions?

Open Problem 2 (Full 3D). Does the Phi-renormalization technique extend to non-axisymmetric initial data? The Q6 prime-indexed spectral damping operator is proposed as the next step in this program.

These are stated explicitly as open. The present contribution is the Phi-renormalization theorem and the closure of the axisymmetric-with-swirl regularity problem within the Q1-approximation class.

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